Discrete Electronics Project: Class AB Audio Amplifier Design and Simulation Report

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***Abstract*—Analog One develops a discrete audio amplifier that combines preamplification, voltage gain, and a Class AB power stage for an $8\,\Omega$ loudspeaker. The design targets $20\,\mathrm{W}$ of output power and uses commercial component values. Calculated values and simulation results are organized in tables to summarize bias, signal levels, and heat dissipation. The power-stage test delivered $22.8\,\mathrm{W}$ at $10\,\mathrm{kHz}$. The report also identifies the interstage checks needed to assess the complete amplifier.**

***Index Terms*—Audio amplifier, Class AB, discrete circuits, MOSFET, simulation.**

I. Project Description

The amplifier prepares an incoming audio signal, increases its voltage, and delivers current to a loudspeaker. A BJT preamplifier provides initial signal conditioning, a MOSFET voltage amplifier establishes the power-stage input level, and a complementary Class AB output stage drives the load. The design considers signal transfer, stable bias, and component heating. Each stage is evaluated separately before the complete signal path is assessed. A common $\pm 24\,\mathrm{V}$ supply serves the system: the preamplifier uses the positive rail and ground, while the voltage and power stages use both rails.

II. Block Diagrams

A. Overall Amplifier

The three stages divide signal conditioning, voltage gain, and loudspeaker drive into distinct functions.

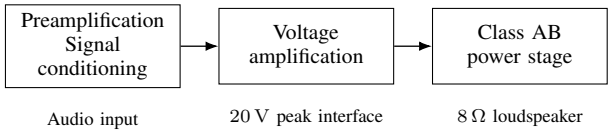


Figure 1. Main stages of the audio amplifier.

B. Preamplification Stage

The common-emitter stage increases the input signal level. Emitter feedback helps stabilize its gain, while coupling capacitors keep the DC bias separate from the adjacent stages.

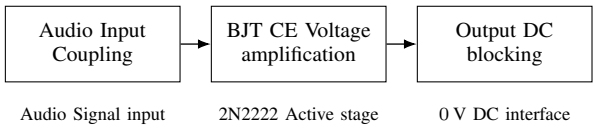


Figure 2. Functional blocks of the BJT preamplifier stage.

C. Voltage Amplification Stage

Two common-source stages provide voltage gain while preserving the signal polarity overall. An active load supplies current to the second stage.

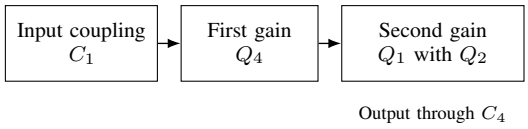


Figure 3. Signal path through the voltage amplification stage.

D. Class AB Power Stage

Class AB operation was selected for its smooth transition between the positive and negative portions of the signal and moderate resting power consumption. The output stage supplies loudspeaker current while retaining most of the input voltage amplitude [1, secs. 7–2 and 9–2].

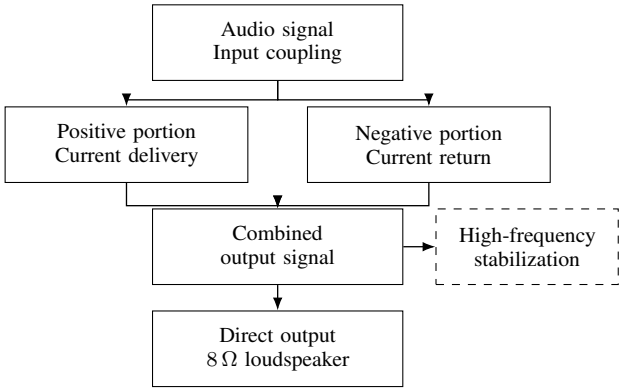


Figure 4. Functional blocks of the selected Class AB output arrangement.

III. Design Considerations and Calculated Values

A. Preamplification Stage

The preamplifier uses a common-emitter 2N2222 with an unbypassed emitter resistor. Its bias network uses the positive supply rail and ground. Table I summarizes the nominal operating point of this configuration.

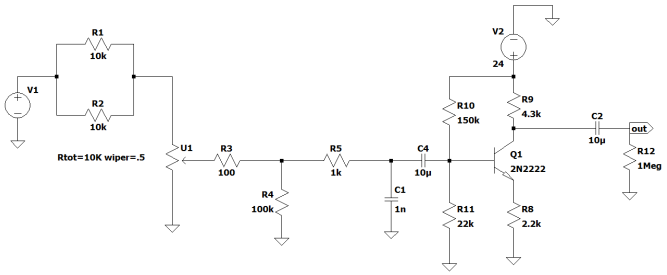


Figure 5. 2N2222 preamplifier test circuit.

Table I
Calculated Preamplifier Design Values

Quantity	Value
Base-network equivalent resistance	$19.19\,\mathrm{k\Omega}$
Base-network open-circuit voltage	$3.07\,\mathrm{V}$
Base current	$6.744\,\mu\mathrm{A}$
Base voltage	$2.94\,\mathrm{V}$
Emitter voltage	$2.24\,\mathrm{V}$
Collector voltage	$19.65\,\mathrm{V}$
Collector current	$1.012\,\mathrm{mA}$
Emitter current	$1.018\,\mathrm{mA}$
Collector-to-emitter voltage	$17.41\,\mathrm{V}$
Dynamic emitter resistance	$25.55\,\Omega$
Nominal unloaded voltage gain	$-1.93\,\mathrm{V/V}$
Transistor resting loss	$17.61\,\mathrm{mW}$
Estimated junction temperature	$28.52\,^{\circ}\mathrm{C}$

Nominal calculation uses the $+24\,\mathrm{V}$ rail, ground, a current-gain parameter of 150, a base-to-emitter drop of $0.70\,\mathrm{V}$, and $25\,^{\circ}\mathrm{C}$ ambient.

B. Voltage Amplification Stage

The voltage stage establishes the signal level required by the power stage. Table II summarizes its calculated interface values. Bias estimates include the recorded resting drain voltages.

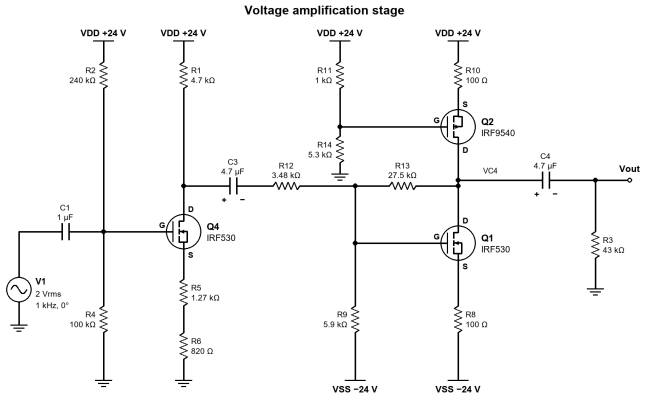


Figure 6. MOSFET voltage amplifier with an active load and a $43\,\mathrm{k\Omega}$ test load.

Table II
Calculated Voltage-Stage Interface Values

Quantity	Value
Required output, peak	$20.00\,\mathrm{V}$
Required output, RMS	$14.14\,\mathrm{V}$
Required voltage gain	$7.07\,\mathrm{V/V}$
Input resistance	$70.6\,\mathrm{k\Omega}$
Input coupling corner	$2.25\,\mathrm{Hz}$
Load current at target, RMS	$0.329\,\mathrm{mA}$
Load power at target	$4.65\,\mathrm{mW}$
First-stage source resistance	$2.09\,\mathrm{k\Omega}$
DC feedback-path current	$0.715\,\mathrm{mA}$
Active-load bias-divider current	$3.81\,\mathrm{mA}$
R11 resting loss	$14.5\,\mathrm{mW}$
R14 resting loss	$76.9\,\mathrm{mW}$
Q4 gate-to-source voltage	$3.90\,\mathrm{V}$
Q4 drain-to-source voltage	$13.73\,\mathrm{V}$

Table III
Calculated Voltage-Stage Bias and Resting Losses

Device	Gate (V)	Source (V)	Current (mA)	Loss (mW)
Q4	7.06	3.16	1.51	20.8
Q1	−19.78	−23.69	3.07	72
Q2	20.19	23.62	3.79	90

Q1 and Q2 entries are nominal resting estimates.

C. Class AB Power Stage

The complementary source followers supply current directly to the loudspeaker. Adjustable gate bias sets a small resting drain current and keeps the output near zero DC.

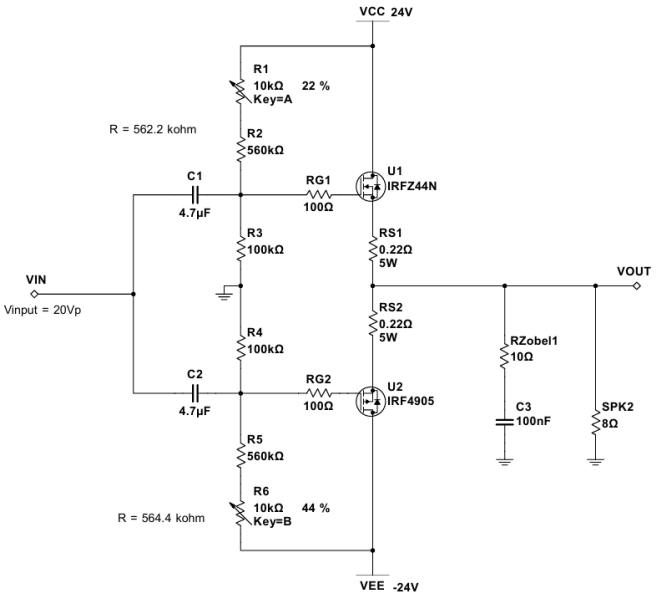


Figure 7. Class AB power-stage schematic with IRFZ44N and IRF4905 MOSFETs.

Table IV
CALCULATED POWER-STAGE DESIGN VALUES

Quantity	Value
Output-power target	20 W
Required load voltage, RMS	12.65 V
Required load voltage, peak	17.89 V
Required load voltage, peak-to-peak	35.78 V
Required load current, RMS	1.58 A
Required load current, peak	2.24 A
Upper gate bias, initial	+3.6243 V
Lower gate bias, initial	−3.6123 V
Upper effective bias resistance	562.2 kΩ
Lower effective bias resistance	564.4 kΩ
R1 effective setting	2.2 kΩ (22%)
R6 effective setting	4.4 kΩ (44%)
Quiescent drain current, I_{DQ}	50 mA per device
Resting drop across RS1 and RS2	11 mV each
Output DC target	0 V
Input resistance per branch	84.9 kΩ
Input coupling corner	0.40 Hz
Resting loss per source resistor	0.55 mW

The initial gate settings are adjusted through the zero-input DC analysis to meet the resting-current and output-DC targets.

1) Heat Dissipation: The thermal assessment uses an ambient temperature of 25 °C for all components. Each output transistor uses a separate NTE495 aluminum heat sink for natural-convection cooling [2].

Table V
POWER-STAGE LOSSES AND CALCULATED TEMPERATURES

Component	Power used	Temperature (°C)
U1	6.58 W	123.7
U2	6.63 W	119.5
R1	4.28 µW	25.0
R2	1.09 mW	25.2
R3, R4	2.13 mW	25.3
R5	1.08 mW	25.2
R6	8.50 µW	25.0
RG1	0.171 mW	25.0
RG2	0.913 mW	25.1
RS1, RS2	0.732 W	65.3
RZobel1	0.0714 W	30.4
C1	0.0590 mW	25.0
C2	0.311 mW	25.1
C3	0.0284 W	57.9

Temperatures are analytical estimates. U1/U2 entries refer to the junction; passive entries refer to the body. Grouped values apply to each component. RS1/RS2 use a sizing allowance; U1/U2 powers are recorded test losses.

Table VI
CALCULATED OUTPUT-TRANSISTOR THERMAL CONDITIONS

Condition	Loss per device	U1 (°C)	U2 (°C)
Resting target	1.20 W	43.0	42.1
Thermal stress check	10 W	175.0	167.5

Estimates use a constant-resistance model: one NTE495 per device at 13 °C/W and a 0.5 °C/W mounting interface. The heat-sink rating is specified at a 75 °C rise in natural convection [2]. Both MOSFETs have a maximum junction temperature of 175 °C [3], [4]; the 10 W check leaves no thermal margin for U1.

IV. COMPONENT SELECTION CRITERIA

A. Preamplification Stage

The bias divider sets the transistor operating point, while the emitter resistor provides local feedback. Coupling capacitors isolate the DC levels of adjacent stages.

Table VII
PREAMPLIFIER COMPONENTS

Reference	Value	Function
Q1	2N2222	Common-emitter amplifier
R10	150 kΩ	Upper bias resistor
R11	22 kΩ	Lower bias resistor
R9	4.3 kΩ	Collector resistor
R8	2.2 kΩ	Emitter feedback
C4, C2	10 µF each	Input and output coupling
R12	1 MΩ	Output return
C1	1 nF	High-frequency filtering
R3	100 Ω	Input series resistor
R4	100 kΩ	Input return
R5	1 kΩ	Filter resistor
U1	10 kΩ	Input volume control

B. Voltage Amplification Stage

IRF530 devices provide voltage gain, while an IRF9540 acts as the active load [5], [6]. The input and feedback resistances set the gain and loading between the stages.

Table VIII
VOLTAGE-STAGE COMPONENTS

Reference	Value	Function
Q_4, Q_1	IRF530	Common-source gain
Q_2	IRF9540	Active load
R_2, R_4	240 kΩ, 100 kΩ	Q_4 gate bias
R_1	4.7 kΩ	Q_4 drain load
R_5, R_6	1.27 kΩ, 820 Ω	Q_4 source path
R_{12}, R_{13}	3.48 kΩ, 27.5 kΩ	Input and feedback
R_9	5.9 kΩ	Return to -24 V
R_8, R_{10}	100 Ω each	Source resistors
R_{11}, R_{14}	1 kΩ, 5.3 kΩ	Q_2 gate bias
R_3	43 kΩ	Output test load
C_1	1 μF	Nonpolar input coupling
C_3, C_4	4.7 μF each	Interstage and output

C. Class AB Power Stage

The complementary MOSFETs share the load current. Source resistors provide bias-current checks, and the output RC network supports high-frequency operation [3], [4].

Table IX
POWER-STAGE COMPONENTS AND SELECTION RATINGS

Reference	Value or model	Selection rating
U1	IRFZ44N	N-channel MOSFET
U2	IRF4905	P-channel MOSFET
R2, R5	560 kΩ	1%, 0.25 W
R3, R4	100 kΩ	1%, 0.25 W
R1, R6	10 kΩ	Trimmers, 0.25 W
RG1, RG2	100 Ω	0.25 W
RS1, RS2	0.22 Ω	5 W
C1, C2	4.7 μF	50 V, electrolytic
RZobel1	10 Ω	2 W
C3	100 nF	50 V, ceramic, code 104
SPK2	8 Ω	30 W nominal rating
HS1, HS2	NTE495	Two heat sinks, one per MOSFET

1) Heat-Sink Recommendations: The NTE495 supplied by Teltron is selected for both output devices. Table X also lists two compact local alternatives. Their thermal resistance must be confirmed before either replaces the selected model.

Table X
COMPACT HEAT-SINK OPTIONS IN COSTA RICA

Model and supplier	Size (mm)	Thermal information
NTE495, Teltron	$29.9 \times 25.4 \times 12.7$	13 °C/W; selected [2]
AD73819, MicroJPM	$43 \times 17 \times 30$	Rating not published [7]
DEV-CB0214-68, CRCibernética	$50 \times 45 \times 18$	Rating not published [8]

NTE495 data follow the manufacturer's datasheet. Alternative dimensions follow supplier product specifications; manufacturer datasheets are not provided in those listings.

V. SIMULATION METHODOLOGY

DC analysis evaluates resting bias. Transient tests measure signal levels and power, while frequency sweeps and harmonic analysis assess signal transfer and distortion. LTspice and Multisim were used for the preamplifier; the voltage and power stages were evaluated in Multisim.

Table XI
STAGE TEST CONDITIONS

Stage	Signal	Load or condition
Preamplifier	1 kHz transient	DC and AC sweep
Voltage amplifier	2 V RMS, 1 kHz	43 kΩ
Power amplifier	20 V peak, 10 kHz	8 Ω

All tests use the common ±24 V supply; the preamplifier draws from +24 V and ground. Voltage-stage Fourier analysis includes nine harmonics with a 2 s stop time to allow settling. Power-stage bias is checked with zero audio input.

VI. SIMULATED VALUES AND RESULTS

A. Preamplification Stage

Table XII compares the nominal DC bias with the recorded simulation for this configuration.

Table XII
PREAMPLIFIER DC BIAS RESULTS

Quantity	Calculated	Simulated	Deviation
Base voltage	2.94 V	3.03 V	3.06%
Emitter voltage	2.24 V	2.33 V	4.02%
Collector voltage	19.65 V	19.47 V	0.92%
Collector current	1.012 mA	1.055 mA	4.25%
Collector-to-emitter voltage	17.41 V	17.14 V	1.55%
Output DC	0 V	194.66 pV	—

Table XIII
REPORTED PREAMPLIFIER TRANSIENT OUTPUT

Quantity	Value
Output, peak-to-peak	Approximately 1.0 V
Output, peak	Approximately 0.50 V
Output DC center	0 V

B. Voltage Amplification Stage

The transient test reaches the required output swing. The Fourier results summarize the fundamental and harmonic content after the coupling capacitors have settled.

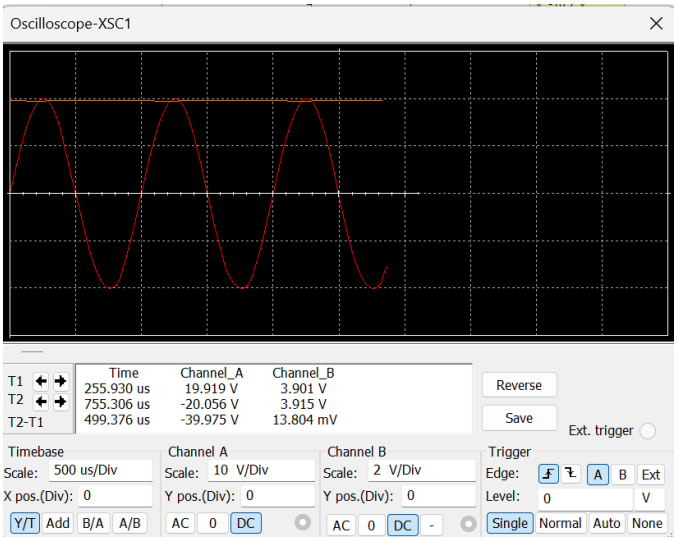


Figure 8. Voltage-stage transient test at 1 kHz; cursors mark opposite output peaks.

Table XIV
VOLTAGE-STAGE SIMULATION RESULTS

Quantity	Value
Q4 drain at rest	16.89 V
Internal output at rest, VC4	−0.125 V
Positive output peak	19.919 V
Negative output peak	−20.056 V
Output DC, Fourier analysis	0.996 mV
Fundamental amplitude	20.112 V peak
Second harmonic	0.077 V peak
Third harmonic	0.193 V peak
THD, through ninth harmonic	1.183%
Small-signal gain near 20 Hz	6.930 V/V
Small-signal gain near 1 kHz	7.118 V/V
Derived output swing	39.975 V peak-to-peak
Derived transient gain	7.07 V/V
Fundamental, RMS equivalent	14.22 V
Output-swing target deviation	Less than 0.07%

C. Class AB Power Stage

The recorded test delivers more than the target load power. The transistor losses represent heat in the output devices and are separate from the power delivered to the loudspeaker.

Table XV
POWER-STAGE VOLTAGES AT 10 kHz

Location	Peak-to-peak	RMS	DC mean
IRFZ44N gate	39.9 V	14.6 V	+3.62 V
IRF4905 gate	39.8 V	14.6 V	−3.62 V
Output AC	38.2 V	13.5 V	—

The output DC mean is not included in this test record. Its adjustment target appears in Table IV.

Table XVI
POWER-STAGE CURRENT AND POWER RESULTS

Quantity	Value
Load current, peak-to-peak	4.78 A
Load current, RMS	1.69 A
Average load power, simulated	22.8 W
U1 average loss, simulated	6.58 W
U2 average loss, simulated	6.63 W
Load power derived from RMS voltage	22.78 W
Load current derived from RMS voltage	1.69 A
Output peak derived from voltage swing	19.1 V
Output-to-input amplitude ratio	95.5%
Power above design target	14%

VII. ANALYSIS AND INTEGRATION

A. Preamplification Stage

The calculated and simulated DC values place the transistor in its active region. Coupling capacitors isolate this bias from the next stage. The interstage check measures signal gain with the voltage amplifier connected.

B. Voltage Amplification Stage

The final resistor network balances the required output swing and harmonic distortion. Its standalone load test establishes the voltage available to the next stage; the connected power-stage input adds loading that must be included in the interface check.

C. Class AB Power Stage

The load-voltage and current readings support the recorded output power. The amplitude ratio is consistent with a source follower, and the recorded transistor losses define the heat-sink operating point. The preceding stage must sustain the required voltage under the connected input load.

VIII. CONCLUSIONS AND NEXT STEPS

A. Preamplification Stage

The BJT stage provides initial voltage gain and DC isolation using the positive supply rail. Loaded interstage measurements complete its integration into the signal path.

B. Voltage Amplification Stage

The MOSFET voltage stage reaches the required signal swing with controlled harmonic distortion in the standalone test. The next check is its output with the power-stage input connected.

C. Class AB Power Stage

The complementary output stage meets the power objective in the recorded test. Its thermal assessment uses one NTE495 heat sink per output transistor at the recorded losses. Loaded interstage measurements, frequency response, and signal-quality checks complete the amplifier assessment.

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